

Q1. Compute the sum of all natural numbers**CODE**

```
total = 0
for i in range(1,500):
    if i % 3 == 0 or i % 5 == 0:
        total = total + i
    print(total)
```

OUTPUT

```
3
8
14
23
33
45
60
78
98
119
143
168
195
225
258
293
329
368
408
450
495
543
593
644
698
753
810
870
933
998
1064
1133
1203
1275
1350
1428
1508
1589
1673
1758
1845
1935
2028
2123
2219
2318
2418
2520
2625
2733
2843
2954
3068
3183
3300
3420
3543
3668
3794
3923
4053
4185
4320
4458
4598
4739
4883
5028
5175
5325
5478
5633
5789
5948
6108
6270
6435
6603
6773
6944
7118
```



Q2. Calculate natural numbers.

CODE

```
sum1 = sum(i**2 for i in range(1, 201))
sum2 = sum(range(1, 201))**2
print(sum2 - sum1)
```

OUTPUT

```
401323300
```

8/26/26, 9:15 PM

Student Submissions — Class Work (22.8.26)

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Q3. Triangle pyramid

CODE

```
a = [[3],[4,6],[2,7,6],[8,5,9,1]]
for i in range(len(a)-2,-1,-1):
    for j in range(len(a[i])):
        a[i][j]+=max(a[i+1][j],a[i+1][j+1])
    print("maximum sum =",a[0][0])
```

OUTPUT

```
maximum sum = 3
maximum sum = 3
maximum sum = 3
maximum sum = 3
maximum sum = 3
maximum sum = 25
```

Q4. Finds the length of the longest sequence of consecutive numbers

CODE

```
a = list(map(int,input().split()))
s = set(a)
m = 0
for x in s:
    if x-1 not in s:
        n = x
        while n in s:
            n+= 1
            m = max(m, n-x)
        print(m)
```

INPUT

```
100 4 200 1 3 2
```

OUTPUT

```
1
2
3
4
4
4
```

Q5. 2D matrix

CODE

```
r, c = map(int, input().split())
a = [list(map(int, input().split())) for _ in range(r)]

for i in range(r):
    for j in range(c):
        if a[i][j] == 0:
            for k in range(c):
                a[i][k] = 0
            for k in range(r):
                a[k][j] = 0

for row in a:
    print(*row)
```

INPUT

```
3 3
1 2 3
4 0 6
7 8 9
```

OUTPUT

```
1 0 0
0 0 0
0 0 0
```